

Crowded Carry in G10 FX and Crypto

Crash risk in FX and crypto carry trades

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Carry pays because it crashes

Our question: do carry trades crash when they are crowded?

- We test it in two markets with the same economics: G10 currencies and crypto.
- Strategy A is carry in seven G10 currencies against the US dollar, with a crowding filter built from CFTC data.
- Strategy C is funding carry in twelve crypto perpetual futures on Binance, including the coins that collapsed, Terra and FTX Token.

Why carry crashes

- Carry borrows the low-yield asset and holds the high-yield one. UIP says this earns nothing, but in the data it earns a premium, with rare large crashes.
- Crashes are worst when the trade is crowded and everyone unwinds at once (Brunnermeier, Nagel & Pedersen 2008).

Hypotheses, fixed before testing

- A: carry earns a premium, crowded months are followed by weaker carry, and it loses in volatility spikes.
- C: funding carry earns a premium with little BTC exposure, and its crashes are coin-specific.

A: G10 FX

C: Crypto

Carry signal	3-month rate differential	Perpetual funding rate
Crowding data	CFTC speculator positions	Funding = leveraged demand
Universe	7 currencies vs USD	12 coins (listed below)
Frequency	Monthly, 1999–2026	Weekly, 2020–2026

Currencies: Japanese yen (JPY), euro (EUR), British pound (GBP), Swiss franc (CHF), Canadian dollar (CAD), Australian dollar (AUD) and New Zealand dollar (NZD), each against the US dollar.

Coins: Bitcoin (BTC), Ethereum (ETH), BNB, XRP, Cardano (ADA), Dogecoin (DOGE), Solana (SOL), Litecoin (LTC), Chainlink (LINK), Avalanche (AVAX), Terra (LUNA) and FTX Token (FTT).

How the two strategies trade

Each step uses only data available at the time of the trade

Strategy A: G10 FX carry, monthly

- 1 Month-end: rank**
Compute each currency's 3-month rate minus the US rate.
- 2 Trade**
Buy the 2 highest-rate currencies and sell the 2 lowest, 50% each, so there is no net dollar bet.
- 3 Crowding check**
Z-score CFTC speculator positions over 36 months; if crowding is above 1, halve all positions.
- 4 Hold one month**
Earn the currency move plus the interest-rate gap.
- 5 Rebalance**
At the next month-end, rank again and swap, paying 3 bp per trade.

Strategy C: crypto funding carry, weekly

- 1 Sunday close: rank**
Average each coin's daily funding rate over the past 7 days, using data up to the day before.
- 2 Trade**
Short the third of coins with the highest funding and buy the lowest third (usually 3 each), so the book is dollar-neutral.
- 3 Hold 7 days**
Earn the price move plus funding; a short receives funding when it is positive.
- 4 Rebalance**
Next Sunday, rank again and swap, paying 5 bp per trade.

Data and methodology

Public data; a research design chosen to keep the tests honest

Data	Source	Coverage
FX spot, 7 currencies	FRED (Fed H.10)	1999–2026
3-month rates	OECD via FRED	1999–2026
Speculator positions	CFTC Commitments of Traders	1986–2026
Crypto price, volume, funding	Binance public archive	2020–2026
Carry factor (validation)	Lustig–Roussanov–Verdelhan	1983–2021

Data fix: before Aug 2000 CFTC lists the same currency futures under 'International Monetary Market'. Matching both names extended positioning back to 1986.

- All parameters were fixed before the first backtest.
- The out-of-sample period starts in 2011 for A and in 2024 for C.
- Costs are 3 bp per unit traded in FX and 5 bp in crypto.
- There is no look-ahead: weights set at t earn the return of $t+1$, and CFTC data are used only after their Friday release.
- There is no survivorship bias, because Terra (LUNA) and FTX Token (FTT) stay in the sample.
- Robustness grids are reported, but they were never used for tuning.

Strategy A: crowding-filtered G10 carry

Yen, euro, pound, Swiss franc, Canadian, Australian and New Zealand dollar vs USD, monthly

1

Carry portfolio

At each month-end we buy the 2 highest-rate currencies and sell the 2 lowest, with equal weights and no net dollar bet.

2

Crowding signal

Each currency's CFTC net speculative position over open interest is z-scored over 36 months.

3

Portfolio crowding

It is the average z of the long leg minus the average z of the short leg.

4

Filter

If crowding is above 1, all positions are cut to half for the next month.

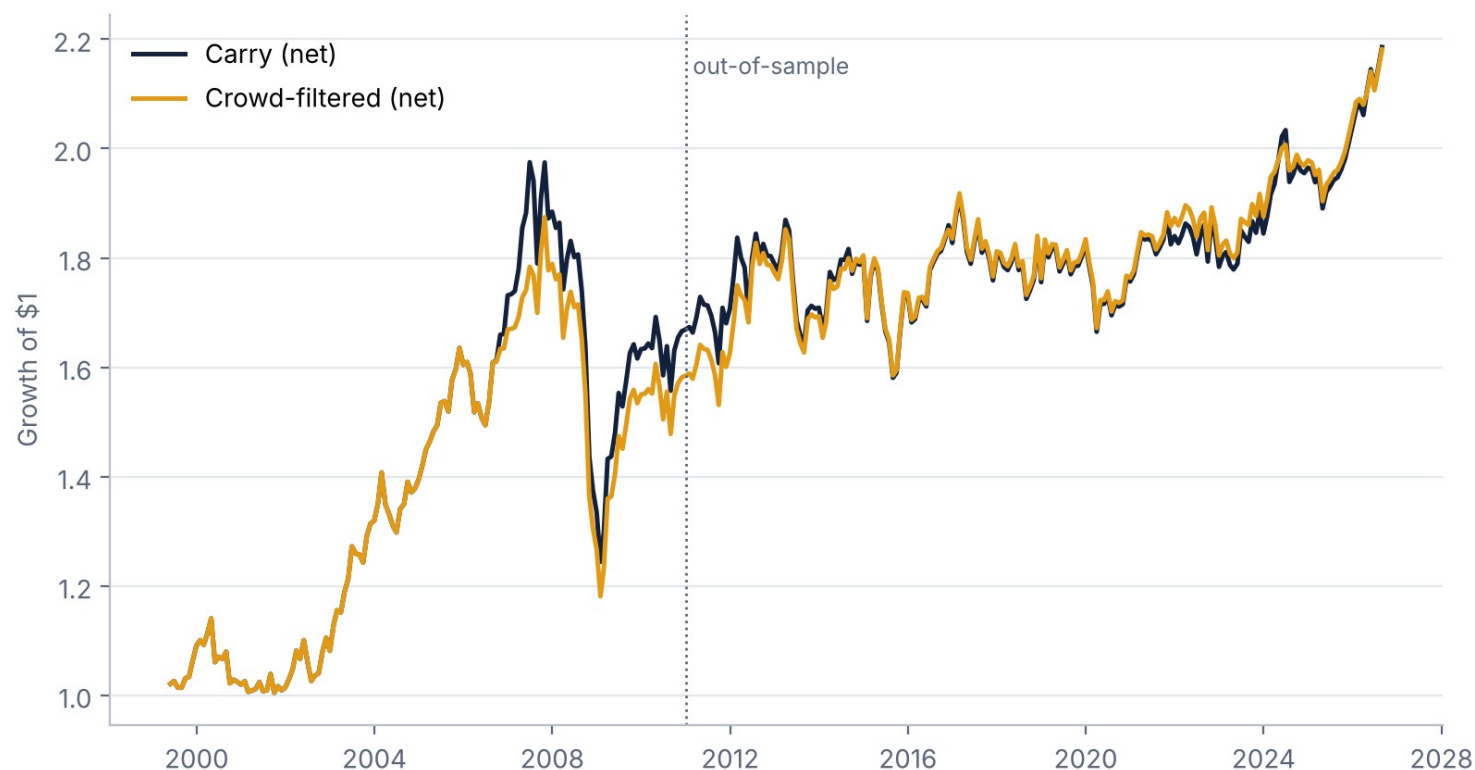
33

crowded months out of 328, 1999–2026

Who we usually hold

- Long: New Zealand dollar (94% of months) and Australian dollar (75%).
- Short: Swiss franc (99%), Japanese yen (52%) and euro (40%).

Crowding filter helps out of sample, not in 2008



0.27 → 0.32

Out-of-sample Sharpe, net: carry vs crowd-filtered (2011–2026)

0.30% vs 0.02%

Next-month carry after non-crowded vs crowded months

0.82

Correlation with the academic LRV carry factor (1999–2021)

Net, 1999–2026	Net return a year	Volatility a year	Max drawdown
Carry	3.3%	8.9%	-37%
Crowd-filtered	3.2%	8.5%	-37%

Strategy A loses when volatility spikes

Monthly net carry on risk factors, Newey-West t-stats, 1999–2021

-1.55

Beta to equity-volatility changes (t = -3.7)

0.35

Beta to the dollar factor (t = 4.1)

0.25%

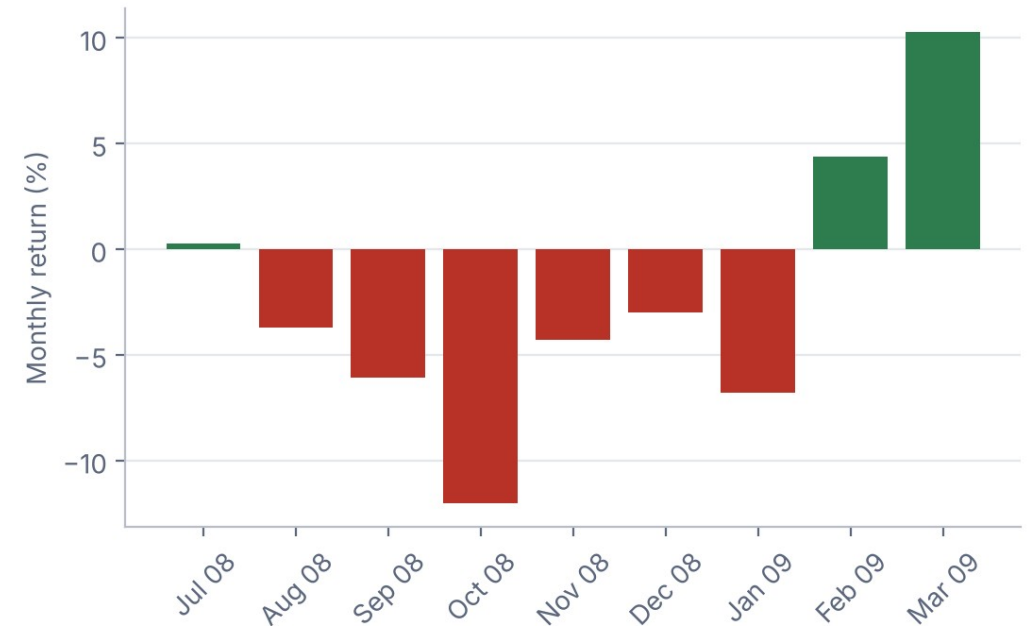
Monthly alpha (t = 1.7): not significant

0.18

R² of the two-factor model

Worst months: Oct 2008, Aug 2007, May 2000, Jan 2009, and the Jan 2015 Swiss franc de-peg.

The 2008 carry crash: -20.5% (Jul 2008 – Mar 2009)



Strategy C: crypto funding carry

Binance perps on BTC, ETH, BNB, XRP, ADA, DOGE, SOL, LTC, LINK, AVAX, LUNA (Terra), FTX; weekly, 2020–2026



- Funding is paid every 8 hours, and positive funding means longs pay shorts.
- Each Sunday we short the third of coins with the highest 7-day funding and buy the lowest third.
- The book is dollar-neutral and pays 5 bp per trade.

0.14

Net Sharpe, full

0.81

Net Sharpe, 2024–26

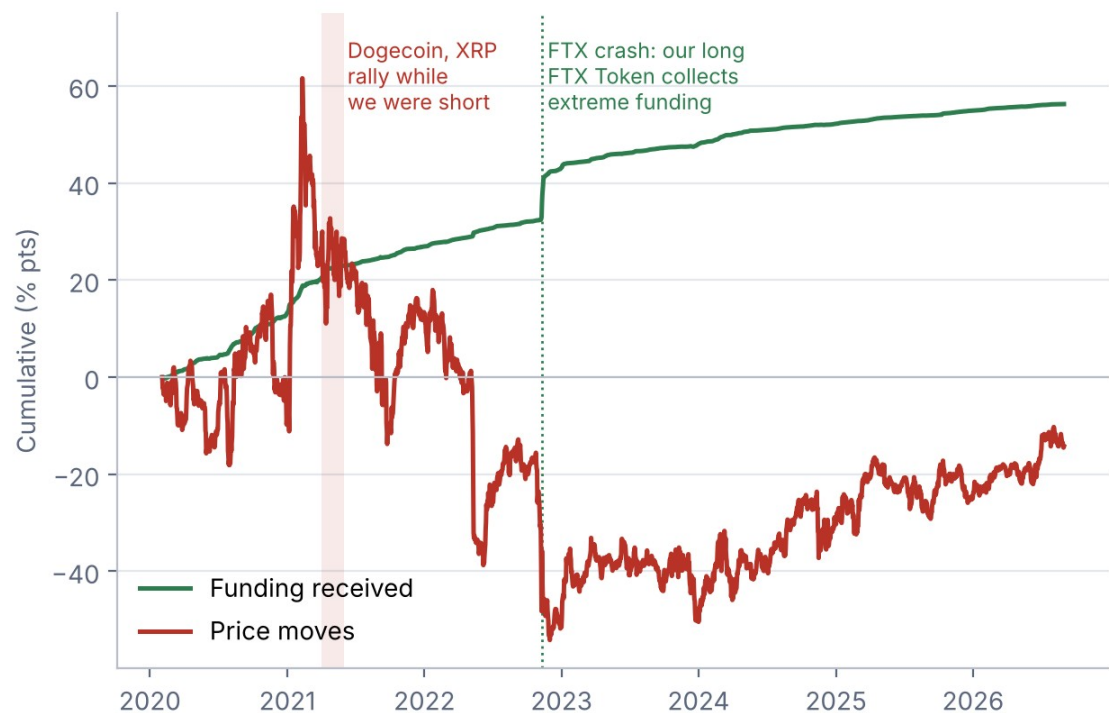
-27%

Worst week: LUNA (Terra) collapse, May 2022

Net of costs	Net return a year	Volatility a year	Max drawdown
Full sample 2020–26	4.1%	30%	-67%
Since 2024	13.9%	17%	-13%

Funding is earned, but price moves dominate

Where the return comes from, and what drives the losses



Total P&L by coin, % pts: Solana +102, BNB +51, Avalanche +36; Terra -87, XRP -58, Dogecoin -43, FTX Token -13.

Robustness: all 6 lookback and rebalancing settings have a positive out-of-sample Sharpe (0.41 to 1.29); our pre-set choice is in the bottom half.

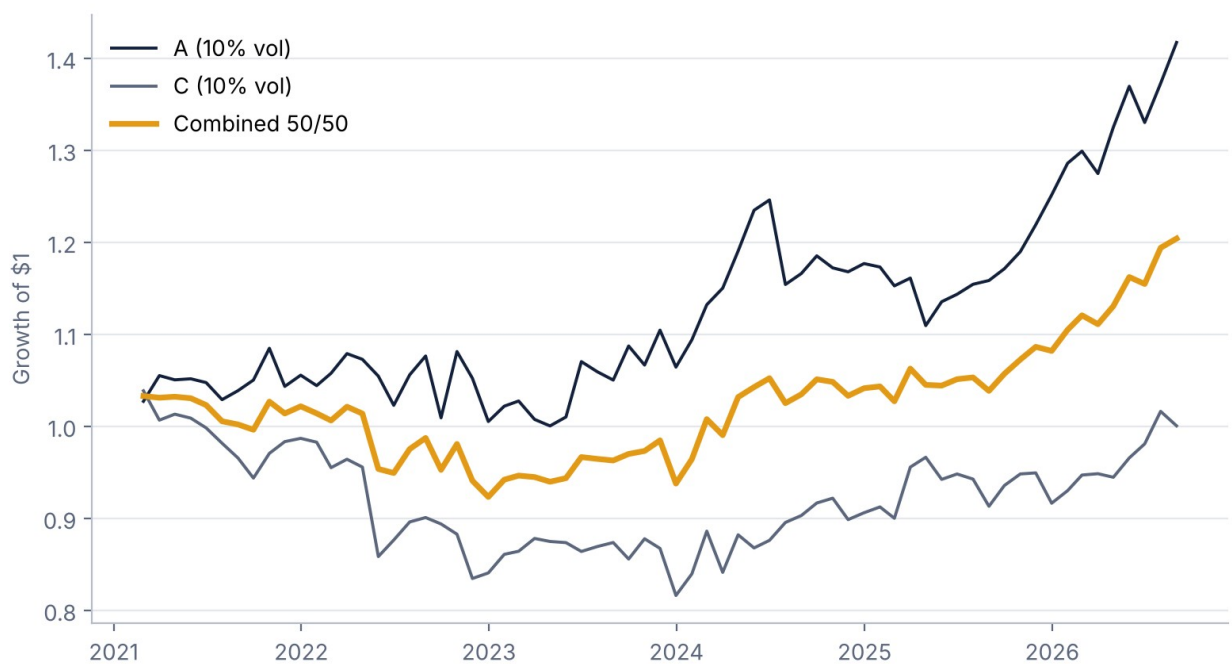
% per year	Funding	Price	Net
Full	+8.5	-2.2	+4.1
In-sample	+12.3	-12.5	-2.5
Out-of-sample	+3.1	+13.0	+13.9

Takeaways

- Funding income is steady at about 8.5% a year, but price moves decide the result.
- The strategy has no market exposure, since its beta to Bitcoin is -0.02.
- Its losses come from single coins: Terra lost 87 points, XRP 58 and Dogecoin 43.
- Our pre-set risk control lowered the Sharpe from 0.14 to -0.16, so it did not help.
- Deeply negative funding signals crowded shorts, not a cheap coin.

Combining FX and crypto carry

Each scaled to 10% vol with lagged volatility, 50/50, 2021-2026



	Sharpe	Vol (%)	Max DD (%)	Worst month (%)
A	0.71	9.4	-11	-7.4
C	0.05	9.3	-21	-10.2
Combined 50/50	0.51	7.0	-11	-5.9

0.11

Correlation between A and C, monthly, 2021-2026

Takeaways

- FX carry and crypto carry are almost uncorrelated, so they crash at different times.
- The 50/50 mix has 7.0% volatility, compared with about 9% for each strategy alone.
- Its worst month is -5.9%, smaller than for FX (-7.4%) or crypto (-10.2%).
- The mix does not beat FX alone on Sharpe (0.51 vs 0.71), because crypto carry was weak in this window.

Conclusions and limitations

Our question: do carry trades crash when they are crowded?

Short answer: partly yes. Crowding makes carry weaker, but it doesn't predict the biggest crashes.

- FX: carry crashes are compensation for risk, and crowding is a modest warning sign, not a reliable crash predictor.
- Crypto: carry also earns a premium and also crashes, but the crashes come from single coins, mostly being long a coin that traders are crowding to short, so crowding tells you where the danger is rather than when it will hit.

Hypotheses: verdict and evidence

Hypothesis	Verdict	Evidence
A1: G10 carry earns a premium	Supported	Net Sharpe 0.37; 0.82 correlation with LRV
A2: crowding predicts weaker carry	Leaning yes	0.02% vs 0.30% next month; OOS Sharpe 0.27 → 0.32; missed 2008
A3: carry loses in volatility spikes	Supported	Vol beta -1.55 (t = -3.7); alpha not significant
C1: funding carry earns a premium	Mixed	Funding +8.5% a year, but net Sharpe 0.14; price moves dominate
C2: no BTC exposure; coin-specific crashes	Supported	BTC beta ≈ 0; Terra -87 pts; worst week -27%
Pre-set risk control helps	Not supported	Sharpe 0.14 → -0.16
FX and crypto carry diversify	Supported	Correlation 0.11; vol 7.0% vs about 9%

What is new

- We use CFTC speculator positioning as a crowding filter on G10 carry.
- We treat crypto funding as the crypto interest-rate gap and as a crowding signal, and test it with the coins that died.

Limitations

- The crypto history is short (2020–2026) and covers only 12 coins.
- There are only 33 crowded FX months, and the 2008 crash was not flagged.
- Costs are simple estimates, and we ignore funding-rate caps, borrow limits and exchange risk.